

Circles and circle theorems - Set 2 - Solutions

Specification MM3.2, MM3.3, M5.8, M5.9.

Answer key

Question	Answer	Topic
1	D	Circles and circle theorems
2	A	Circles and circle theorems
3	B	Circles and circle theorems
4	F	Circles and circle theorems
5	C	Circles and circle theorems

Worked solutions

Question 1 CIR-02 · Circles and circle theorems · spec MM3.2

Find the complete set of values of k for which $x^2 + y^2 + 2kx - 4y + 3k + 4 = 0$ represents a circle.

Answer: D

$(x + k)^2 + (y - 2)^2 = k^2 + 4 - 3k - 4 = k^2 - 3k = k(k - 3)$. A circle needs a positive right-hand side: $k < 0$ or $k > 3$.

Full method - how and why

An equation of circle form only is a circle when the resulting r^2 is positive. That gives an inequality in k .

- 1 Complete both squares: $(x + k)^2 - k^2 + (y - 2)^2 - 4 + 3k + 4 = 0$.

why: Treat k as a constant: half of $2k$ is k , half of -4 is -2 .

- 2 Rearrange: $(x + k)^2 + (y - 2)^2 = k^2 - 3k$.

why: The -4 and $+4$ cancel, leaving $r^2 = k^2 - 3k$.

- 3 A genuine circle needs $r^2 > 0$: $k^2 - 3k > 0$, i.e. $k(k - 3) > 0$.

why: If $r^2 = 0$ the 'circle' is a single point; if $r^2 < 0$ there are no real points at all.

- 4 Solve the quadratic inequality: $k < 0$ or $k > 3$.

why: Roots 0 and 3, upward parabola, positive outside the roots.

Where it goes wrong

- Choosing $0 < k < 3$ - that is where r^2 is negative and there is no circle.
- Forgetting the sign change when moving $-k^2$ across.

Takeaway Complete the square and demand $r^2 > 0$; the shape only exists when the right-hand side is positive.

Question 2 CIR-05 · Circles and circle theorems · spec MM3.2

The line $y = x + c$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 8$. Find the possible values of c .

Answer: A

The distance from $(1, 2)$ to $x - y + c = 0$ is $\frac{|1 - 2 + c|}{\sqrt{2}} = \sqrt{8} = 2\sqrt{2}$, so $|c - 1| = 4$: $c = 5$ or $c = -3$.

Full method - how and why

Tangency means the perpendicular distance from the centre to the line equals the radius. Faster than substituting and using the discriminant.

- 1 Identify centre $(1, 2)$ and radius $\sqrt{8} = 2\sqrt{2}$.

why: Read them straight from the centre-radius form.

- 2 Write the line in the form $x - y + c = 0$ and apply the distance formula: $\frac{|1 - 2 + c|}{\sqrt{1^2 + (-1)^2}} = \frac{|c - 1|}{\sqrt{2}}$.

why: Rearranging $y = x + c$ to $x - y + c = 0$ gives $a = 1$, $b = -1$.

- 3 Set it equal to the radius: $\frac{|c - 1|}{\sqrt{2}} = 2\sqrt{2}$, so $|c - 1| = 4$.

why: Multiply both sides by $\sqrt{2}$: $2\sqrt{2} \times \sqrt{2} = 4$.

- 4 Solve the modulus: $c - 1 = \pm 4$, so $c = 5$ or $c = -3$.

why: Two tangents with that gradient exist - one on each side of the circle.

Where it goes wrong

- Dropping the modulus and finding only $c = 5$.
- Using $r = 8$ instead of $\sqrt{8}$.

Takeaway Tangency $\Leftrightarrow$ distance from centre to line = radius; the modulus always gives two answers.

Question 3 CIR-09 · Circles and circle theorems · spec MM3.2

Find the length of the tangent from the point (7, 1) to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$.

Answer: B

Centre (1, 2), radius 3. Distance from (7, 1) to centre: $\sqrt{36 + 1} = \sqrt{37}$. Tangent length = $\sqrt{37 - 9} = \sqrt{28}$.

Full method - how and why

The tangent, radius and line to the external point form a right-angled triangle. Pythagoras gives the tangent length.

- 1 Put the circle in centre-radius form: $(x - 1)^2 + (y - 2)^2 = 4 + 1 + 4 = 9$, so centre (1, 2), radius 3.

why: Complete the square in x and y; the constant -4 moves across.

- 2 Distance from (7, 1) to the centre: $\sqrt{(7 - 1)^2 + (1 - 2)^2} = \sqrt{36 + 1} = \sqrt{37}$.

why: This is the hypotenuse of the right-angled triangle (the tangent meets the radius at 90°).

- 3 Tangent length = $\sqrt{PC^2 - r^2} = \sqrt{37 - 9} = \sqrt{28}$.

why: Subtract the squares, not the lengths - the right angle is at the point of contact.

Where it goes wrong

- Answering $\sqrt{37}$ (that is the distance to the centre) or $\sqrt{37} - 3$.
- Getting $r = 9$ instead of 3 from the completed square.

Takeaway Tangent length from P: $\sqrt{PC^2 - r^2}$, because tangent $\perp$ radius makes a right-angled triangle.

Question 4 CIR-11 · Circles and circle theorems · spec MM3.2

Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point (5, 0) lies on both circles.

Answer: F

Radii 5 and 3, centres 8 apart = 5 + 3, so the circles touch externally at one point (I true, II false). That point is (5, 0): $25 = 25$ and $9 + 0 = 9$ (III true).

Full method - how and why

Two circles: compare the distance between centres with the sum and difference of the radii.

- 1 Read off centres and radii: (0, 0) with $r = 5$; (8, 0) with $r = 3$.

why: Both are already in centre-radius form.

- 2 Distance between centres: 8. Compare with $r_1 + r_2 = 8$.

why: Equal to the sum means the circles touch externally - exactly one common point.

- 3 I is therefore true, and II (two intersection points) is false.

why: Two intersections would need $|r_1 - r_2| < d < r_1 + r_2$, i.e. $2 < d < 8$ strictly.

- 4 III: check (5, 0) in both: $25 + 0 = 25$ ✓ and $(5 - 8)^2 + 0 = 9$ ✓. **True.**

why: The touching point lies on the line joining the centres, 5 from one and 3 from the other.

- 5 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming circles that 'meet' must meet twice.
- Forgetting to test III in **both** equations.

Takeaway $d = r_1 + r_2$: touch externally; $d = |r_1 - r_2|$: touch internally; strictly between: two points.

Question 5 CIR-16 · Circles and circle theorems · spec MM3.2

The point $P(x, y)$ moves so that its distance from $(3, 0)$ is twice its distance from $(0, 0)$. The locus of P is a circle. Find its radius.

Answer: C

$$(x - 3)^2 + y^2 = 4(x^2 + y^2) \Rightarrow 3x^2 + 3y^2 + 6x - 9 = 0 \Rightarrow x^2 + 2x + y^2 = 3 \Rightarrow (x + 1)^2 + y^2 = 4. \text{ Radius } 2.$$

Full method - how and why

A locus defined by a distance ratio. Write the condition algebraically, square to remove roots, and complete the square.

1 Write the condition: $\sqrt{(x - 3)^2 + y^2} = 2\sqrt{x^2 + y^2}$.

why: 'Distance from A is twice the distance from B' - write both distances with the formula.

2 Square both sides: $(x - 3)^2 + y^2 = 4(x^2 + y^2)$.

why: Squaring is safe because both sides are non-negative; and it removes both roots at once. Note the right-hand side gains a factor 4, not 2.

3 Expand and collect: $x^2 - 6x + 9 + y^2 = 4x^2 + 4y^2 \Rightarrow 3x^2 + 3y^2 + 6x - 9 = 0$.

why: Bring everything to one side.

4 Divide by 3 and complete the square: $x^2 + 2x + y^2 = 3 \Rightarrow (x + 1)^2 + y^2 = 4$, so $r = 2$.

why: Dividing by the common factor first makes the coefficients of x^2 and y^2 equal to 1 - necessary before completing the square.

Where it goes wrong

- Squaring the right side as $2(x^2 + y^2)$ instead of $4(x^2 + y^2)$.
- Forgetting to divide by 3 before completing the square, which corrupts the radius.

Takeaway Distance-ratio loci (Apollonius circles): square the condition, divide so x^2 and y^2 have coefficient 1, then complete the square.